KNOWLEDGE ENGINEERING AND ONTOLOGIES COURSE PROJECT REPORT

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Executive Summary (2 pts)

This project presents the design and implementation of the University Club and Event Management Ontology developed for the Knowledge Engineering and Ontologies course. The main objective of the project is to create a machine-understandable representation of university clubs, students, events, memberships, attendance records, recommendations, and semantic question-answering concepts using Semantic Web technologies. The ontology was developed using OWL and RDF standards and incorporates ontology reuse principles through the integration of FOAF and Schema.org concepts. A knowledge graph was constructed by combining the ontology schema with instance-level data representing students, clubs, events, interests, announcements, and recommendations. The resulting knowledge graph was deployed in GraphDB and queried using SPARQL to retrieve information about memberships, event participation, recommendations, interests, and club activities. Data quality and consistency were supported through SHACL validation rules that define structural constraints for the main ontology entities. The project also explores the integration of Large Language Models (LLMs) with knowledge graphs to support semantic question answering and recommendation workflows. The developed solution demonstrates how ontology engineering, knowledge graph construction, semantic querying, validation technologies, and AI-assisted approaches can be combined to build an extensible and reusable knowledge-based system for university club and event management. The project provides a foundation for future intelligent applications that support personalized event recommendations and knowledge-driven decision making.

Description of the project (2,5 pts)

The objective of this project is to develop a semantic knowledge management system for university clubs and events by applying ontology engineering and Semantic Web technologies. University clubs organize various activities, events, workshops, and social programs; however, information about students, memberships, interests, event participation, and recommendations is often distributed across different systems and represented in an unstructured manner. This project addresses this problem by creating an OWL-based ontology and a knowledge graph that provide a structured, reusable, and machine-understandable representation of university club activities. The ontology was developed using ontology engineering principles and METHONTOLOGY-inspired modeling practices. Existing semantic resources such as FOAF and Schema.org were reused and extended where appropriate. Data was modeled as RDF triples and integrated into a knowledge graph deployed in GraphDB. SPARQL was used to query and analyze the graph, while SHACL was employed to define validation constraints for ensuring data consistency and quality. The project also investigates how Large Language Models can be integrated with the knowledge graph to support semantic question answering and recommendation generation by translating natural language questions into graph-based queries.

The system was developed using Protégé, GraphDB, RDF, OWL, Turtle, SPARQL, SHACL, Widoco, and GitHub Pages. The primary target users are university students, club managers, and university administrators who need efficient access to club and event information. The expected outputs include a reusable ontology, a knowledge graph, SPARQL query implementations, SHACL validation rules, ontology documentation, and a conceptual LLM-assisted semantic question-answering workflow.

Competency Questions

1. Which students are members of a specific club?
2. Which events are organized by a specific club?
3. Which students attended a specific event?
4. Which events match a student's interests?
5. Which events are recommended for a specific student?
6. Which clubs organize technology-related events?
7. How many events does each club organize?
8. Which announcements are associated with a specific event?

Ontology Design (20 pts)

The University Club and Event Management Ontology was developed to represent the entities, relationships, and activities associated with university clubs and events. The ontology was modeled using OWL and RDF standards and follows ontology engineering principles that emphasize clarity, reusability, extensibility, and semantic interoperability. The ontology captures information about students, clubs, events, memberships, attendance records, interests, recommendations, announcements, and semantic question-answering concepts.

Main Classes

The ontology contains the following primary classes:

- Student
- Club
- Event
- Membership
- Attendance
- Location
- EventCategory
- Organizer
- Announcement
- Interest
- Role
- Recommendation
- SemanticQuestion

These classes represent the fundamental concepts required to model university club activities and event management processes.

Object Properties

Object properties define relationships between ontology entities. The main object properties are:

- memberOf (Student → Club)
- organizes (Club → Event)
- attends (Student → Event)
- hasLocation (Event → Location)
- hasCategory (Event → EventCategory)
- hasOrganizer (Event → Organizer)
- hasAnnouncement (Event → Announcement)
- hasInterest (Student → Interest)
- matchesInterest (Event → Interest)
- hasRole (Student → Role)
- recommendedFor (Recommendation → Student)
- recommendsEvent (Recommendation → Event)

These relationships enable semantic navigation across the knowledge graph and support advanced querying and recommendation scenarios.

Data Properties

Data properties are used to store literal values associated with ontology entities. The main data properties include:

- studentId
- clubName
- eventTitle
- eventDate
- attendanceStatus
- announcementText
- interestName
- roleName
- questionText

These properties provide descriptive information required for querying and knowledge graph population.

TBox (Terminological Knowledge)

The TBox defines the conceptual schema of the ontology. It includes all classes, object properties, data properties, domains, ranges, subclass relationships, and semantic constraints. Several ontology reuse principles were applied. The Student class was modeled as a subclass of foaf from the FOAF ontology, while Event and Location were modeled as subclasses of schema and schema from Schema.org. This approach improves interoperability with existing Semantic Web standards and promotes ontology reuse.

ABox (Assertional Knowledge)

The ABox contains instance-level data representing real examples from the domain. Example individuals include:

- ArdaStudent
- AyseStudent
- MehmetStudent
- ZeynepStudent
- CanStudent
- ComputerScienceClub
- RoboticsClub
- MusicClub
- EntrepreneurshipClub
- SportsClub
- AIWorkshop2026
- PythonBootcamp2026
- RoboticsMeetup2026
- MusicNight2026
- StartupSeminar2026
- CampusRun2026

These individuals are connected through RDF triples and form the basis of the knowledge graph used in GraphDB.

Modeling Decisions

Several important modeling decisions were made during ontology development. Students, clubs, events, and interests were modeled as classes because they represent reusable conceptual categories. Individual students, specific clubs, and concrete events were modeled as instances belonging to these classes. Roles such as ClubMember and ClubPresident were represented separately using the Role class rather than creating subclasses of Student. This approach allows a student to have multiple roles and increases flexibility. Recommendations were modeled as independent entities to support future recommendation systems and LLM-assisted applications. SemanticQuestion was introduced to represent natural language questions that can be mapped to SPARQL queries and knowledge graph operations.

The ontology also distinguishes clearly between conceptual knowledge and instance-level data. This separation simplifies maintenance, enables ontology reuse, and supports future extensions such as semantic reasoning, recommendation engines, and intelligent question-answering systems.

Data Acquisiton (10 pts)

The data used in this project was collected through a combination of manually created university club management scenarios and information derived from publicly available university club structures. Since the primary objective of the project is ontology development and knowledge graph construction rather than large-scale data analytics, a representative dataset was created to demonstrate the functionality of the ontology and semantic technologies.

The dataset includes information about students, clubs, events, interests, locations, announcements, recommendations, and participation relationships.

Data Sources

The data sources used in this project include:

- University club management scenarios designed for ontology population
- Publicly available examples of university club activities and event announcements
- Existing Semantic Web vocabularies, including FOAF and Schema.org, which were reused during ontology design
- Synthetic example data created to represent realistic students, clubs, and events

The project did not rely on external APIs or large public datasets because the focus was on demonstrating ontology engineering concepts and knowledge graph construction techniques.

Data Format

Initially, the information was represented in a conceptual tabular form similar to CSV-style records describing students, clubs, events, and relationships. During ontology population, the data was transformed into RDF/Turtle format. The final knowledge graph data was stored in Turtle (.ttl) files and imported into GraphDB for querying and analysis.

Data Collection Methodology

The data collection process followed several steps:

1. Identification of the main entities required by the ontology.
2. Definition of representative examples for students, clubs, events, interests, and recommendations.
3. Creation of relationships between entities according to the ontology schema.
4. Conversion of the collected information into RDF triples.
5. Deployment of the resulting data into GraphDB.

This approach ensured that the dataset adequately represented the competency questions and semantic relationships required by the project.

Data Preprocessing

Before transforming the data into RDF format, several preprocessing steps were applied:

- Removal of duplicate entities and relationships
- Standardization of entity names and identifiers
- Consistent naming of clubs, events, and interests
- Verification of relationships between students, clubs, and events
- Validation of date values and textual descriptions
- Transformation of the data into RDF-compatible structures

The preprocessing stage improved consistency and reduced the possibility of semantic ambiguity within the knowledge graph.

RDF Mapping

The processed data was mapped to ontology concepts and properties. Examples include:

- Student records → Student class
- Club information → Club class
- Event descriptions → Event class
- Interest areas → Interest class
- Announcements → Announcement class
- Membership relations → memberOf property
- Event participation → attends property
- Recommendations → Recommendation class and recommendation properties

After mapping, the information was represented as RDF triples using the subject–predicate–object structure and loaded into GraphDB.

Data Quality and Limitations

The dataset is sufficient for demonstrating ontology modeling, knowledge graph construction, SPARQL querying, and SHACL validation. However, several limitations exist. The dataset is relatively small and was partially generated for educational purposes rather than collected from a real production environment. As a result, it may not capture all complexities of large-scale university club ecosystems. Future work could involve collecting real-world university club data through APIs, institutional databases, or web scraping techniques to expand the coverage and realism of the knowledge graph.

Knowledge Graph Construction (20 pts)

The knowledge graph was constructed by combining the ontology schema with instance-level data represented in RDF format. The ontology defines the conceptual structure of the domain, while the instance data populates the ontology with concrete examples of students, clubs, events, interests, recommendations, and announcements. Together, these components form a semantic knowledge graph that can be queried, validated, and extended.

RDF Model

The knowledge graph follows the RDF (Resource Description Framework) data model. RDF represents knowledge as triples consisting of three components:

- Subject
- Predicate
- Object

The subject represents a resource, the predicate represents a property or relationship, and the object represents either another resource or a literal value.

The general RDF structure can be expressed as:

Subject → Predicate → Object

For example:

ArdaStudent → memberOf → ComputerScienceClub

indicates that ArdaStudent is a member of the Computer Science Club.

Representative RDF Triples

The following examples illustrate RDF triples used in the knowledge graph:

ArdaStudent rdf:type Student

ArdaStudent memberOf ComputerScienceClub

ArdaStudent attends AIWorkshop2026

ComputerScienceClub organizes AIWorkshop2026

AIWorkshop2026 hasLocation RoomA101

AIWorkshop2026 hasCategory TechnologyCategory

RecommendationForArdaAI recommendedFor ArdaStudent

RecommendationForArdaAI recommendsEvent AIWorkshop2026

These triples establish semantic connections between entities and enable complex graph-based queries.

Namespace Definitions

The ontology uses several namespaces to support interoperability and ontology reuse:

uce: http://example.org/university-club-event-ontology#

owl: http://www.w3.org/2002/07/owl#

rdf: http://www.w3.org/1999/02/22-rdf-syntax-ns#

rdfs: http://www.w3.org/2000/01/rdf-schema#

xsd: http://www.w3.org/2001/XMLSchema#

foaf: http://xmlns.com/foaf/0.1/

schema: http://schema.org/

The FOAF and Schema.org vocabularies were reused to improve semantic interoperability and ontology reusability.

Knowledge Graph Deployment

The ontology schema and instance data were deployed using GraphDB. A repository named “university-club-event-kg” was created and configured using the OWL2-RL reasoning profile. Two RDF/Turtle datasets were imported into the repository:

1. university-club-event-ontology-v2.ttl
2. club-event-data.ttl

The ontology file contains the conceptual schema (TBox), while the data file contains the instance-level knowledge (ABox). After importing both datasets, GraphDB automatically stored the information as RDF triples and made it available for SPARQL querying.

The resulting knowledge graph contains students, clubs, events, announcements, interests, recommendations, and semantic question entities connected through meaningful semantic relationships.

Graph Structure

The knowledge graph contains multiple interconnected entity types. Students are linked to clubs through membership relationships, clubs organize events, events are associated with categories and locations, and recommendations connect students with relevant events based on their interests. These relationships create a connected graph structure that supports semantic querying and knowledge discovery.

Screenshots of the GraphDB repository, imported datasets, and query results are included in the appendix to illustrate the deployed knowledge graph and its structure.

SPARQL Queries (10 pts)

SPARQL queries were implemented to demonstrate the functionality of the University Club and Event Management Knowledge Graph. The queries were designed according to the competency questions defined earlier in the report. They cover basic retrieval, semantic relationship traversal, recommendation retrieval, and aggregation.

Query 1: List All Students

Type: Basic retrieval

Related Competency Question: Which students exist in the knowledge graph?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?student ?label

WHERE {

?student a uce:Student .

OPTIONAL { ?student rdfs:label ?label . }

}
```

Purpose: This query retrieves all student individuals stored in the knowledge graph.

Expected Output: ArdaStudent, AyseStudent, MehmetStudent, ZeynepStudent, and CanStudent.

Query 2: List All Clubs

Type: Basic retrieval

Related Competency Question: Which clubs are represented in the knowledge graph?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>

SELECT ?club ?clubName

WHERE {

?club a uce:Club ;

uce:clubName ?clubName .

}
```

Purpose: This query lists all university clubs and their names.

Expected Output: Computer Science Club, Robotics Club, Music Club, Entrepreneurship Club, and Sports Club.

Query 3: List All Events with Dates and Locations

Type: Basic retrieval

Related Competency Question: Where and when does a specific event take place?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
SELECT ?event ?eventTitle ?eventDate ?location
WHERE {
?event a uce:Event ;
uce:eventTitle ?eventTitle ;
uce:eventDate ?eventDate ;
uce:hasLocation ?location .
}
```

Purpose: This query retrieves all events with their titles, dates, and locations.

Expected Output: AI Workshop 2026, Python Bootcamp 2026, Robotics Meetup 2026, Music Night 2026, Startup Seminar 2026, and Campus Run 2026 with their corresponding dates and locations.

Query 4: Find Events Organized by the Computer Science Club

Type: Semantic relationship query

Related Competency Question: Which events are organized by a specific club?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
SELECT ?event ?eventTitle
WHERE {
uce:ComputerScienceClub uce:organizes ?event .
?event uce:eventTitle ?eventTitle .
}
```

Purpose: This query follows the organizes relationship from the Computer Science Club to its events.

Expected Output: AI Workshop 2026 and Python Bootcamp 2026.

Query 5: Find Students and Their Club Memberships

Type: Semantic relationship query

Related Competency Question: Which students are members of a specific club?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?student ?studentLabel ?club ?clubName
WHERE {
?student a uce:Student ;
uce:memberOf ?club .
```

```
?club uce:clubName ?clubName .
OPTIONAL { ?student rdfs:label ?studentLabel . }
}
```

Purpose: This query retrieves the relationship between students and the clubs they belong to.

Expected Output: Arda Student → Computer Science Club, Ayşe Student → Robotics Club, Mehmet Student → Music Club, Zeynep Student → Entrepreneurship Club, and Can Student → Sports Club.

Query 6: Find Students and Their Interests

Type: Semantic relationship query

Related Competency Question: Which interests are associated with each student?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?student ?studentLabel ?interest ?interestName
WHERE {
?student a uce:Student ;
uce:hasInterest ?interest .
?interest uce:interestName ?interestName .
OPTIONAL { ?student rdfs:label ?studentLabel . }
}
```

Purpose: This query retrieves the interests connected to each student.

Expected Output: Arda Student → Artificial Intelligence and Software Development, Ayşe Student → Robotics and Artificial Intelligence, Mehmet Student → Music, Zeynep Student → Entrepreneurship and Software Development, and Can Student → Sports.

Query 7: Find Events Matching Artificial Intelligence Interest

Type: Reasoning / semantic matching query

Related Competency Question: Which events match a student's interests?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
SELECT ?event ?eventTitle
WHERE {
?event a uce:Event ;
uce:matchesInterest uce:ArtificialIntelligence ;
uce:eventTitle ?eventTitle .
}
```

Purpose: This query identifies events that semantically match the Artificial Intelligence interest.

Expected Output: AI Workshop 2026.

Query 8: Find Recommended Events for Arda Student

Type: Recommendation query

Related Competency Question: Which events are recommended for a specific student?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
SELECT ?recommendation ?event ?eventTitle
WHERE {
?recommendation a uce:Recommendation ;
uce:recommendedFor uce:ArdaStudent ;
uce:recommendsEvent ?event .
?event uce:eventTitle ?eventTitle .
}
```

Purpose: This query retrieves the events recommended for Arda Student.

Expected Output: AI Workshop 2026 and Python Bootcamp 2026.

Query 9: Count How Many Events Each Club Organizes

Type: Aggregation query

Related Competency Question: How many events does each club organize?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
SELECT ?club ?clubName (COUNT(?event) AS ?eventCount)
WHERE {
?club a uce:Club ;
uce:clubName ?clubName ;
uce:organizes ?event .
}
GROUP BY ?club ?clubName
ORDER BY DESC(?eventCount)
```

Purpose: This query counts the number of events organized by each club.

Expected Output: Computer Science Club organizes two events, while Robotics Club, Music Club, Entrepreneurship Club, and Sports Club organize one event each.

Query 10: List Event Announcements

Type: Basic retrieval / semantic relationship query

Related Competency Question: Which announcements are associated with a specific event?

```
PREFIX uce: <http://example.org/university-club-event-ontology#>
SELECT ?eventTitle ?announcementText
WHERE {
?event a uce:Event ;
```

```
uce:eventTitle ?eventTitle ;
uce:hasAnnouncement ?announcement .
?announcement uce:announcementText ?announcementText .
}
```

Purpose: This query retrieves event titles and their announcement texts.

Expected Output: Announcement texts for AI Workshop 2026, Python Bootcamp 2026, Robotics Meetup 2026, Music Night 2026, Startup Seminar 2026, and Campus Run 2026.

Validation (10 pts)

SHACL (Shapes Constraint Language) was used to define validation rules for checking the consistency and quality of the University Club and Event Management Knowledge Graph. The purpose of using SHACL is to ensure that important entities in the graph contain the required properties, correct datatypes, and valid relationships. The validation rules were written in a separate file named shacl-validation.ttl and imported into GraphDB together with the ontology and instance data.

The SHACL validation file defines several NodeShapes for the main ontology classes, including Student, Club, Event, Announcement, Interest, and Recommendation. Each shape specifies structural requirements that individuals of the target class should satisfy.

Student Shape

The StudentShape targets all individuals of the Student class. It requires each student to have exactly one student ID, at least one club membership, and at least one interest.

Example constraints:

```
uce:StudentShape
  a sh:NodeShape ;
  sh:targetClass uce:Student ;
  sh:property [
    sh:path uce:studentId ;
    sh:datatype xsd:string ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:message "Each student must have exactly one student ID." ;
  ] ;
  sh:property [
    sh:path uce:memberOf ;
    sh:class uce:Club ;
    sh:minCount 1 ;
    sh:message "Each student should be a member of at least one club." ;
  ] .
```

This constraint ensures that student instances are identifiable and properly connected to clubs.

Club Shape

The ClubShape validates club individuals. It requires each club to have exactly one club name and to organize at least one event.

Example constraint:

```
uce:ClubShape
  a sh:NodeShape ;
  sh:targetClass uce:Club ;
  sh:property [
    sh:path uce:clubName ;
    sh:datatype xsd:string ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:message "Each club must have exactly one club name." ;
  ] .
```

This ensures that club resources are clearly named and usable in SPARQL query results.

Event Shape

The EventShape is used to validate event instances. It requires each event to have an event title, event date, location, category, and announcement.

Example constraints:

```
uce:EventShape
  a sh:NodeShape ;
  sh:targetClass uce:Event ;
  sh:property [
    sh:path uce:eventTitle ;
    sh:datatype xsd:string ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:message "Each event must have exactly one event title." ;
  ] ;
  sh:property [
    sh:path uce:eventDate ;
    sh:datatype xsd:date ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:message "Each event must have exactly one event date." ;
  ] .
```

These constraints ensure that events contain the minimum descriptive information required for searching, filtering, and recommendation.

Recommendation Shape

The RecommendationShape validates recommendation instances. It requires each recommendation to be connected to exactly one student and exactly one event.

Example constraints:

```
uce:RecommendationShape
  a sh:NodeShape ;
  sh:targetClass uce:Recommendation ;
  sh:property [
    sh:path uce:recommendedFor ;
    sh:class uce:Student ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:message "Each recommendation must be assigned to exactly one student." ;
  ] ;
  sh:property [
    sh:path uce:recommendsEvent ;
    sh:class uce:Event ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:message "Each recommendation must recommend exactly one event." ;
  ] .
```

This prevents incomplete recommendation records and supports reliable recommendation queries.

Validation Results

The SHACL file was imported into GraphDB as part of the validation layer. The knowledge graph was designed to satisfy the defined constraints. No intentional invalid records were included in the final dataset. Therefore, the expected validation result is that the graph conforms to the defined shapes. If violations occur in future versions, they would likely be caused by missing required properties such as event titles, student IDs, club names, or recommendation links. These issues could be corrected by adding the missing RDF triples or revising inconsistent datatype values.

Overall, SHACL validation improves the reliability of the knowledge graph by ensuring that the main entities follow consistent structural rules. It also provides a foundation for future data quality control when the ontology is populated with larger real-world university club datasets.

Evaluation, Discussion and Conclusion (10 pts)

The University Club and Event Management Ontology successfully demonstrates the application of ontology engineering, knowledge graph construction, semantic querying, and data validation technologies within a realistic university domain. From a technical perspective, the ontology provides a consistent and structured representation of students, clubs, events, memberships, interests, announcements, and recommendations.

The use of OWL classes, object properties, and data properties enabled the creation of meaningful semantic relationships between entities. The ontology was successfully deployed in GraphDB together with instance-level RDF data, forming a functional knowledge graph that supports SPARQL querying and SHACL-based validation.

The SPARQL queries developed during the project were able to answer the competency questions defined during the ontology design phase. Basic retrieval queries, semantic relationship queries, and aggregation queries all produced the expected results. Furthermore, the SHACL validation rules helped define structural constraints for important entities such as students, clubs, events, and recommendations, improving the overall consistency of the knowledge graph. Since no direct LLM integration was implemented in the current version of the project, the evaluation focuses primarily on the ontology and knowledge graph components rather than AI-based question answering capabilities.

Several limitations should also be acknowledged. The dataset used in the project is relatively small and was partially created for educational purposes. As a result, it does not fully represent the complexity of a real university club management environment. In addition, the ontology currently focuses on a limited set of concepts and relationships. More advanced reasoning rules, larger datasets, and additional semantic constraints could further improve the system. Scalability was not extensively evaluated because the knowledge graph contains only a moderate number of entities and RDF triples.

In conclusion, the project achieved its primary objective of developing a semantic representation of university clubs and events through ontology engineering and knowledge graph technologies. The project provided practical experience with OWL modeling, RDF representation, GraphDB deployment, SPARQL querying, SHACL validation, ontology documentation, and semantic system design. Throughout the project, important concepts related to ontology development, knowledge graph construction, and semantic interoperability were learned and applied in practice. Future work may include integrating real-world university datasets, expanding the ontology with additional concepts, implementing advanced reasoning mechanisms, and exploring natural language interfaces or LLM-assisted semantic question-answering systems.

References and format (2,5 pts)

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Appendix

https://ardaomer.github.io/university-club-event-ontology/widoco/UniversityClubEventOntology/index-en.html

https://github.com/ArdaOmer/university-club-event-ontology